

Yawar Hussain

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Scopus: [URL](#) (Citations-1296, H-Index-19)

Research Focus: Near-surface geophysicist with an extensive landslide-monitoring record, including multi-temporal seismic, electrical, and GPR characterisation of active slopes, slidequake typology, and interferometric deformation tracking, combined with hands-on Distributed Acoustic Sensing (DAS) field deployment on an operational cable, directly transferable to using existing telecom infrastructure for landslide and avalanche monitoring. In the field I can do fibre splicing, OTDR/QC of DAS installations, and I can connect to remote systems through SCADA or the local network. I also work with SQL, databases, and write frontend and backend code.

EMPLOYMENT

Specialist Infrastructure Hazard Monitoring (01/2026 to Present): HydroResearch, Stockholm, Sweden. Supporting the expansion of fibre-optic monitoring technologies (DTS, DSS, DAS) through integrated field measurements, system operation, and data evaluation. Contribute to acoustic monitoring, quality control, alarm follow-up, and technical reporting, ensuring end-to-end DAM and near mine subsidence monitoring from measurement to interpretation.

Research Associate (11/2024 to 06/2025), Department of Sustainable Earth Systems Sciences, University of Texas at Dallas, USA. Engaged in near-surface geophysics and modeling to study the Pecos Valley Aquifer System in Texas. Worked with various versions of the PulseEKKO GPR series, configuring them for both field operations and instructional demonstrations. Responsibilities also included teaching a graduate-level course on structural health monitoring and communication of Earthquake sciences to the students for diverse academic backgrounds. Also completed an advanced level Postdoctoral Teaching Certificate.

Post-doctoral Researcher (01/2023 to 11/2024), Department of Earth Science, 'Sapienza' University of Rome, Italy. Analysis of data from multiparametric monitoring systems for detection of slope instabilities affecting strategic infrastructures and cultural heritage: i) stable-structural stability risk assessment ii) monitoring of Le Capore and Pescheria hydrostructures iii) multi-hazard assessment framework for historical landslide iv) seismic monitoring of Acuto Field Lab. (supervision: Dr. S. Martino).

Visiting Professor (04/2024 to 04/2024), University of Brasilia, Brazil. The activities include application of innovative shallow geophysical techniques, especially GPR to study geomorphological targets in tropical environments. Administration of a three credit hours course on application of geophysical techniques for geomorphological studies for students from department of geography and institute of geosciences.

Post-doctoral Researcher (01/2021 to 12/2022), Department of Geology, University of Liege, Belgium. Geophysical applications for the site selection of the Einstein Telescope installation and mega landslide in el Hierro. Installation of signal-station, mobile and semi-permanent arrays of broadband seismometers (FDSN NETWORK:5B) (supervision: Dr. H.B. Havenith).

Post-doctoral Researcher (01/2020-12/2020), Environmental Engineering and Earth Sciences, Clemson University, USA. The deformation/strain recorded by state-of-the-art strainmeter during an injection test in Oklahoma (supervision: Dr. L. Murdoch).

Post-doctoral Researcher (08/2018-01/2020), National Center for Research and Conservation of Caves–CECAV–Instituto Chico Mendes de Conservação da Biodiversidade–ICMBio, Brasília, Brazil.

Geophysical (GPR and ERT) and geotechnical characterization of subsurface caves for geohazard assessment and their impacts on the vulnerability of the underlying karst aquifer (supervisions: Dr. R. Uagoda & Dr. R. Rebolledo).

SELECTED FIELDWORKS

Distributed Acoustic Sensing (DAS): Installed and operated DAS systems at LKAB's Kiruna mine, Sweden, and led ambient-noise and active-source MASW data acquisition on pre-installed fiber-optic cable at the X3 Dam, Sweden.

Distributed Temperature Sensing (DTS): Participated in fieldworks deploying DTS systems for temperature monitoring at the Boliden and Kiruna mine sites, Sweden.

Seismic Node Deployment: Deployed and operated SmartSolo seismic nodes for both active-source MASW and passive seismic surveys.

Seismological Instrumentation: Installed and maintained Kinometrics Q330 seismic sensors at the ACUTO Field Laboratory, Italy.

Archaeological Geophysics: Carried out geophysical surveys at major archaeological sites in Greece, including Epidaurus, Kalapodi, and Aegina Kolona, as part of the STABLE and Triqutera projects.

Landslide Investigation, San Andrés (Spain): Acquired and interpreted geophysical data to characterize the San Andrés landslide, El Hierro, Canary Islands.

Landslide Investigation, Pays de Herve (Belgium): Led geophysical field investigations of three active paleo-landslides in the Pays de Herve using MASW, electrical resistivity tomography (ERT), and ambient-noise measurements.

Regional Seismological Network Deployment: Installed and monitored seismological networks and mobile seismic arrays across the Belgium, Netherlands, and Germany border region, the candidate site for the Einstein Telescope. [5B \(2021-2022\): ET Database of Belgium](#)

Slope Stability Experiment: Designed and ran a prototype "Normal Slope Experiment" studying plate-loading-induced microfracturing in clay, combining short-period seismometer arrays, ambient-noise interferometry, and terrestrial laser scanning to characterize stress-relief mechanisms.

Landslide Characterization, Sobradinho (Brazil): Investigated the Sobradinho landslides across dry and rainy seasons using dense seismic arrays, ERT, MASW, GPR, and soil sampling.

River Erosion Studies: Designed field experiments combining ambient-noise measurements and GPR to assess a river's erosive potential.

PROJECTS

- Hanging Wall Monitoring by DAS, LKAB mine, Kiruna, Sweden. Design and processing of a Distributed Acoustic Sensing (DAS) survey for hanging wall stability monitoring, combining passive ambient-noise interferometry (dv/v velocity monitoring, resonance-frequency tracking) with active-source MASW dispersion analysis for shear-wave velocity characterisation; authored the feasibility assessment and technical report.
- Dam Stability Monitoring by DAS, X3 Dam, Sweden. Site characterisation and dynamic monitoring of an embankment dam using Silixa DAS interrogators, working alongside geophysicists from Silixa/Luna Innovations. Time-lapse monitoring of the dam crest tracked geophysical indicators of instability, including changes in seismic velocity, changes in resonance frequency, and clustering of microseismic events; results are being compared against piezometric pressure data and other in-situ monitoring datasets.
- Multiscale Geophysical and Petrophysical Monitoring of Coastal Cliffs and Urban Slopes in Northeastern Brazil (submitted 2026, pending decision), CNPq (Brazilian Federal Funding Agency). Study areas: Pipa Beach cliffs (Rio Grande do Norte) and Salvador urban slopes (Bahia). Role:

collaborating researcher, contributing to satellite remote sensing, shallow geophysics (ERT, GPR), passive seismic monitoring (HVSR, ambient noise interferometry, dv/v), and petrophysical characterisation of coastal cliff and urban slope instability.

- Analysis of the impacts generated by seismic activity of natural and artificial origin using records from a broadband seismic station in the city of Arequipa, IBA-IB-71-2020, Funding Agency: UNSA INVESTIGA, Universidad Nacional de San Agustín de Arequipa, Peru.
- CAPESPrint international mobility grant, placed as a visiting professor in University of Brasilia, Brazil.
- Mobility grant to Geosystem Hellas, Greece in framework of STABLE: STructural stABiLity risk assEssment.
- ACEA postdoctoral grant: Analysis of data from multiparametric monitoring systems for slope instability control.
- Wallonia-Brussels International (WBI) Excellence Grant: Geophysical applications for the site selection of the Einstein Telescope installation and mega landslide in El Hierro, Spain.
- CAPES postdoctoral grant: The application of geophysical and geotechnical monitoring technologies contributes to the understanding of the dynamics of mass movements.
- ICMBio and Vale S.A. postdoctoral grant: Use of applied geophysics to support vulnerability assessment of the karst system through geological and conduit mapping.
- PhD grant: Analysis of the dynamics of an unstable artificial slope using Nanoseismic Monitoring and numerical simulation, Funding agency: FAPDF, PI: Hertinez-Carvajal, Hernan, PhD Candidate: Yawar Hussain.

EDUCATION

Ph.D. in Geotechnical Engineering	08/2014–09/2018
Department of Civil and Environmental Engineering, University of Brasilia, Brazil.	
M.Phil in HydroGeophysics	03/2012–03/2014
Department of Earth Sciences, Quaid-i-Azam University, Islamabad, Pakistan.	
Masters in Geophysics	02/2009–02/2011
Department of Earth Sciences, Quaid-i-Azam University, Islamabad, Pakistan.	
Bachelor's Degree	06/2007–08/2009
University of the Punjab, Lahore, Pakistan.	

EDITORIAL CONTRIBUTIONS

- **Handling Editor:** Seismica, (2026–present).
- **Associate Editor:** International Journal of Economic and Environmental Geology (IJEEG) (2026–present).
- **Topic Editor:** (currently accepting papers) Role of Innovative Near Surface Geophysical Techniques in Understanding Geological Instabilities: A Geohazard Perspective, Discover Geosciences, Springer
- **Topic Editor:** (completed) Editorial: Near-surface geophysics in Latin American contexts: its applications, education, and societal perspectives as a whole, Frontiers in Earth Science
- **Review Editor:** Geohazards and Georisk, Frontiers in Earth Science
- **Reviewer:** Geoscience Communication.

SELECTED PUBLICATIONS

Geophysics in Geohazards:

- Xavier, et al. Hussain, et al (2026) Impact Assessment of Infrastructures Exposure to Weak Motion Vibrations in Arequipa, Peru, Brazilian Journal of Geophysics, [URL](#)
- Sokolicek, et al. Hussain, et al (2025) Engineering-geological and geophysical surveys for archaeological risk assessment and previous mitigation measures at the CH-site of Aegina Kolonna, Greece, Italian Journal of Engineering Geology and Environment, [URL](#)
- Innocenti, et al. Hussain et al (2025) Geophysical reconnaissance of the Pays de Herve landslides: case study of three active paleo-slides in a dissected tableland, Landslides, [URL](#)
- Antonielli, et al. Hussain, et al. (2025). Local seismic response in the historical centre of Nafplio (Greece) as a tool for seismic risk management, Natural Hazards, [URL](#)
- Cárdenas-Soto, et al. Hussain et al. (2021) Seismic ambient noise tomography to retrieve near-surface properties in soils with significant 3D lateral heterogeneity. The case of Quinta Colorada building in Chapultepec, Mexico, Natural Hazards, [URL](#)
- Qian, et al. Hussain., et al. (2024). Impact of environmental stressors on jointed rock cliffs by acoustic emission sensing: preliminary findings from the Acuto Field Laboratory (Central Italy), Rock Mechanics and Rock Engineering, [URL](#)
- Hussain, et al. (2022). Review on the Geophysical and UAV-Based Methods Applied to Landslides. Remote Sensing, [URL](#)
- Klimeš, Hussain, et al. (2023). New Insights into the Internal Structures and Geotechnical Rock Properties of the Giant San Andrés Landslide, El Hierro Island, Spain. Remote Sensing, [URL](#)
- Hamza, et al. Hussain, et al. (2019). Geohazard Assessment of Landslides in South Brazil: Case Study. Geotechnical and Geological Engineering, [URL](#)
- Hussain et al. (2019). Multiple Geophysical Techniques for Investigation and Monitoring of Sobradinho Landslide, Brazil. Sustainability, [URL](#)
- Hussain et al. (2019). Monitoring of Sobradinho landslide (Brasília, Brazil) and a prototype vertical slope by time-lapse interferometry. Brazilian Journal of Geology, [URL](#)
- Hussain et al. (2020). Seasonal variation in Rayleigh wave ellipticity as a possible indicator of earthflow mobility: a case study of Sobradinho landslide compared with pile load testing, Earth Sciences Research Journal, [URL](#)
- Hussain et al. (2019). Typological analysis of slidequakes emitted from landslides: experiments on an expander body pile and Sobradinho landslide (Brasilia, Brazil). REM–International Engineering Journal, [URL](#)
- Hamza, et al. Hussain et al. (2022). Evaluation of the subgrade reaction for closely-spaced shallow foundations on sand, Geotechnical Engineering (Proceedings of the ICE), [URL](#)
- Hussain, et al. (2020). Characterization of Sobradinho landslide in fluvial valley using MASW and ERT methods. REM–International Engineering Journal, [URL](#)
- Hussain et al. (2020). Seismic signature of mudflow tremor resulted from Brumadinho (Brazil) tailings dam failure. REM–International Engineering Journal, [URL](#)
- Hussain et al. (2019). Introductory review of potential applications of nanoseismic monitoring in seismic energy characterization. Journal of Engineering Research, [URL](#)
- Condori,, Hussain (2022). Variabilities in urban ambient noise before, during and after COVID-19 lockdown in the Peruvian capital, Brazilian Journal of Geophysics, [URL](#)

- Akhter, et al. Hussain (2018). Petrophysical relationship for density prediction using Vp & Vs in Meyal oilfield, Potwar sub-basin, Pakistan, Geodesy and Geodynamics, [URL](#)

Articles on Karst Studies:

- Gao, Hussain, et al. (2023). A catchment scale GPR application potential zonation: case study from SW China, Environmental Earth Science, [URL](#)
- Hussain, et al. (2020). Detection of cover collapse doline and other epikarst features by multiple geophysical techniques, case study of Tarimba Cave, Brazil. Water, [URL](#)
- Hussain, et al. (2020). The potential use of geophysical methods to identify cavities, sinkholes and pathways for water infiltration. Water, [URL](#)
- Ferreira, Hussain et al. (2023). UAV-based doline mapping in Brazilian karst: a cave heritage protection reconnaissance, Open Geoscience, [URL](#)
- Ferreira, Hussain et al. (2022). A semi-automatic approach for doline mapping in Brazilian covered karst: the way forward to the vulnerability assessment, Acta Carsologica, [URL](#)
- Caldeira, et al. Hussain, (2021). Late Quaternary episodes of clastic sediment deposition in the Tarimba Cave, Central Brazil. Quaternary International, [URL](#)

SELECTED CONFERENCES

- Human-Induced Seismicity Front in Texas and Risk Preparedness: University Students Exercise, 2025 NSF GAGE/SAGE Community Science Workshop, Bloomington, USA (poster)
- Using Geophysics Instruments to Support "Learn by Doing" in University Teaching, 2025 NSF GAGE/SAGE Community Science Workshop, Bloomington, USA (session chair and talk)
- Geophysical investigations to quantify seismic hazard at archeological sites of Epidaurus, Kalapodi and Egina Kolona (Greece), ESC-2024, Corfu, Greece (presentation by S. Martino).
- Geohazard Identification at Le Capore Spring, Central Apennines, Italy, AIGAA Napoli 2024 (talk)
- Passive seismic monitoring of a jointed rock mass at Acuto Field Lab (Italy), Eurock 2025, Norway (presentation by Matteo)
- Multi-scale monitoring of rock mass deformations in a mid-latitude climatic context through a Fiber Bragg Grating array, Eurock 2025, Norway
- Integrated geotechnical, acoustic and microseismic monitoring of a jointed rock-wall prone to block failures: results from the Acuto Field Lab (Italy), EUROCK-2024, Spain
- Integration of single-station ambient seismic noise techniques in multi-hazard assessment framework: case study of the historical landslide of San Vito Romano (Italy), ESC-2024, Corfu, Greece (poster by S. Marano).
- Landslide detection and monitoring by integrating electrical, seismic and interferometric techniques in a multi-hazard perspective: the case of San Vito Romano (RM), GNGTS 2025, Italy
- Slope-scale deformation process interacting with hydrostructure: insights from geotechnical monitoring, geophysical investigation and remotely sensed observations, XIVth International Symposium on Landslides-2024, France.
- Geohazard identification at the Le Capore Spring (Central Apennines, Italy) retrieved from engineering-geological modelling and passive seismic approaches, AIGAA Napoli 2024, Italy.
- The near-surface velocity structures of an incipient volcanic flank collapse revealed by seismic studies, EGU General Assembly-2022, ID-EGU22-10450.

- Preliminary results presentation from the seismic geophysical campaigns (single station and array) in Belgian side of ETEST area, ET - Site Studies and Characterization, 8-11 November 2021, Nuoro, Italy, URL
- Introduction to passive seismic imaging techniques in the E-Test area (Belgium), at Joint ET Industrial Advisory Board - 2021 meeting, Liege, Belgium.
- Site Characterization of Hillslopes in Ribeirão Contagem Fluvial Valley (Brasilia, Brazil) using Geophysical Methods, SAGEEP 2019 - 32nd Annual Symposium on the Application of Geophysics to Engineering and Environmental Problems, USA.
- Detecting the Soil Instabilities under Variable Loading Conditions by Rayleigh Wave Velocity Change, IEEE/OES 2017 RIO ACOUSTICS, Rio de Janeiro, Brazil, URL. A Brief Introduction to Nanoseismic Monitoring, 15th International Congress of the Brazilian Geophysical Society and the EXPOGEf 2017, Rio de Janeiro, Brazil.
- Seasonal Monitoring of Hydrological Stresses Developed by Varying Degree of Rainfall Induced Pore-Pressures Using Noise Data, 15th International Congress of the Brazilian Geophysical Society and the EXPOGEf 2017, Rio de Janeiro, Brazil.
- Thickness and shear wave velocity measurement of sediment package at meteorite impact crater site of Colonia, São Paulo City, Brazil, VII SimBGf SIMPÓSIO BRASILEIRO DE GEOFÍSICA, Ouro Preto, Minas Gerais, Brazil.